

Summer Preview - 6th Grade

Reagan's School Day

Thursday, June 18

School day: 12:00 PM - 11:40 PM

What's in this packet:

- Cover sheet (this page) with today's full schedule
- 7 detail pages (one per block, with notes space)
- 7 worksheets embedded in packet

Today's Schedule

1. 10:00 PM • 10 min

Funny clip

A quick fun video clip to start the afternoon (10 minutes or less).

-> Detail page in this packet (notes space below)

2. 10:10 PM • 30 min

Measurement types + overview

What we measure and the tools we use: length, weight/mass, volume/capacity, temperature, and time. General overview of measurement and why units matter, with both customary (US) and metric examples.

-> Detail page in this packet (notes space below)

3. 10:40 PM • 30 min

Measurement conversion intro

Intro to converting within a system: inches to feet, cups to quarts, and the metric ladder (mm/cm/m, mL/L, g/kg). A few worked examples and easy practice.

-> Detail page in this packet (notes space below)

4. 11:10 PM • 30 min [Science]

Intro: Why Water Rolls Off a Duck (watch first)

WHAT WE'RE DOING TODAY: "Hydro" means WATER. We're learning WHY water rolls off ducks and birds instead of soaking in - then after lunch we test it for real in the Duck Hydro Lab.

=== WATCH (~10 min of video) ===

1) PBS Deep Look - "What Actually Makes Water Roll Off a Duck's Back?" (~4 min): <https://www.pbs.org/video/what-actually-makes-water-roll-off-a-ducks-back-jg5gme/>

2) PBS Kids, Elinor Wonders Why - "Why Do Birds Have Feathers?" (~4 min): <https://pbskids.org/videos/watch/why-do-birds-have-feathers/111517>

=== HOW IT CONNECTS (read together) ===

A feather isn't one solid piece. It has a center stick (shaft) with rows of tiny branches (barbs), and each barb has even tinier hooks (barbules) that zip together like Velcro into a smooth, tight surface. Ducks also have a special oil gland near their tail. When a duck "preens," she spreads that oil all over her feathers. Oil and water don't mix - so the oily, tightly-zipped feather makes water pull into little round beads and roll right off (we call that water-repelling, or "hydrophobic" = water-fearing). That same trick keeps a layer of air trapped underneath, so the duck stays warm AND dry. THAT is the hydro connection: feather structure + preen oil = water beads up and slides off.

=== SEE IT UP CLOSE ===

Labeled feather diagram (shaft, vane, barbs, barbules, hooks): <https://askabiologist.asu.edu/feathers>

Museum of Science "Feather Under a Microscope" zoom: <https://www.mos.org/explore-topics/feathers>

=== TRY IT (hands-on) ===

Hold a real feather. Gently pull the vane apart so it splits, then stroke it back together - it "zips" shut! Those are the barbule hooks.

PREDICT before lunch: When we drip plain water on a feather (and on a paper towel) in the Hydro Lab, what will happen to each? What about after we add a drop of soap?

-> Detail page in this packet (notes space below)

5. 12:00 PM • 60 min

Lunch + reset

Lunch and a calm reset before the afternoon.

-> Detail page in this packet (notes space below)

6. 1:00 PM • 40 min

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3-Duck Measurement Adventure

Hands-on with our 3 live ducks! Measure each duck (length beak-to-tail, height, and weight if we can). Then fill the bathtub or a tub of water and explore volume/capacity: how many cups to fill it, convert cups to mL/L, and compare each duck's measurements. Practice converting length (in to cm) and volume (cups to liters) using real numbers from the ducks.

-> Detail page in this packet (notes space below)

7. 1:40 PM · 30 min [Science]

Duck Hydro Lab: Why Water Beads Off a Duck

Science + birds! Explore water with the ducks: (1) Drip water on a feather vs. a paper towel and watch it bead/roll off - talk about how duck feathers are waterproof thanks to preen oil. (2) Sink-or-float in the water dish: predict, then test small safe objects. (3) Stir in a tiny drop of dish soap and watch water stop beading - that is why soap on feathers is bad. Draw what you saw and write 1 sentence on why ducks stay dry.

-> Detail page in this packet (notes space below)

Packet hash: d4b75b324fb44bd2... · Generated for 2026-06-18 · If anything changes before school, this packet will be re-sent.

1. Funny clip

10:00 PM · 10 min

A quick fun video clip to start the afternoon (10 minutes or less).

LEARNING OBJECTIVES

- Students will identify elements that make a clip funny.
- Students will recall details from the clip.
- Students will express their feelings about the clip.

INSTRUCTIONS

Today, we're going to watch a short, funny video clip! Pay close attention to what happens and what makes you laugh. After watching, you'll answer a few quick questions about the clip. This is just for fun and to get our brains ready for the afternoon!

WORKSHEETS & ACTIVITIES

Funny clip - Practice

1. What was the funniest part of the clip for you?

2. Describe one funny thing a character or animal did.

3. What was the main idea or event happening in the clip?

ANSWER KEY (for adult use only)

1. Answers will vary, e.g., 'when the dog tried to catch the ball'
2. Answers will vary, e.g., 'The cat jumped really high.'
3. Answers will vary, e.g., 'A silly animal trying to do something.'

Block 1 of 7 · Reagan · 2026-06-18

2. Measurement types + overview

10:10 PM · 30 min

What we measure and the tools we use: length, weight/mass, volume/capacity, temperature, and time. General overview of measurement and why units matter, with both customary (US) and metric examples.

LEARNING OBJECTIVES

- Identify different types of measurement.
- Match measurement types to common tools.
- Understand why units are important in measurement.

INSTRUCTIONS

Today, we're exploring different ways to measure things! You'll learn about measuring length, weight, volume, temperature, and time. Think about what we use to measure each one. Complete the practice questions below to show what you've learned about measurement types and tools.

WORKSHEETS & ACTIVITIES

Measurement types + overview - Practice

1. What type of measurement tells us how long something is?

2. What tool do we use to measure temperature?

3. Is a cup used for measuring length, weight, or volume?

4. What unit might you use to measure your weight in the US?

5. What type of measurement tells us how heavy something is?

ANSWER KEY (for adult use only)

1. Length
2. Thermometer
3. Volume
4. Pounds
5. Weight or Mass

Block 2 of 7 · Reagan · 2026-06-18

3. Measurement conversion intro

10:40 PM · 30 min

Intro to converting within a system: inches to feet, cups to quarts, and the metric ladder (mm/cm/m, mL/L, g/kg). A few worked examples and easy practice.

LEARNING OBJECTIVES

- Students will be able to convert inches to feet.
- Students will be able to convert cups to quarts.
- Students will be able to convert between common metric units (mm, cm, m).

INSTRUCTIONS

Today, we'll learn about converting measurements! This means changing a measurement from one unit to another, like inches to feet. Remember that 12 inches make 1 foot, and 4 cups make 1 quart. For metric units, think about how many small units make a larger one, like 100 centimeters in 1 meter. Use these facts to help you solve the problems below.

WORKSHEETS & ACTIVITIES

Measurement conversion intro - Practice

1. How many feet are in 24 inches?

2. How many quarts are in 8 cups?

3. Convert 300 centimeters to meters.

4. How many inches are in 3 feet?

ANSWER KEY (for adult use only)

1. 2 feet
2. 2 quarts
3. 3 meters
4. 36 inches

Block 3 of 7 · Reagan · 2026-06-18

4. Intro: Why Water Rolls Off a Duck (watch first)

11:10 PM · 30 min · Science

WHAT WE'RE DOING TODAY: "Hydro" means WATER. We're learning WHY water rolls off ducks and birds instead of soaking in - then after lunch we test it for real in the Duck Hydro Lab.

=== WATCH (~10 min of video) ===

- 1) PBS Deep Look - "What Actually Makes Water Roll Off a Duck's Back?" (~4 min): <https://www.pbs.org/video/what-actually-makes-water-roll-off-a-ducks-back-jg5gme/>
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=== HOW IT CONNECTS (read together) ===

A feather isn't one solid piece. It has a center stick (shaft) with rows of tiny branches (barbs), and each barb has even tinier hooks (barbules) that zip together like Velcro into a smooth, tight surface. Ducks also have a special oil gland near their tail. When a duck "preens," she spreads that oil all over her feathers. Oil and water don't mix - so the oily, tightly-zipped feather makes water pull into little round beads and roll right off (we call that water-repelling, or "hydrophobic" = water-fearing). That same trick keeps a layer of air trapped underneath, so the duck stays warm AND dry. THAT is the hydro connection: feather structure + preen oil = water beads up and slides off.

=== SEE IT UP CLOSE ===

Labeled feather diagram (shaft, vane, barbs, barbules, hooks): <https://askabiologist.asu.edu/feathers>
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=== TRY IT (hands-on) ===

Hold a real feather. Gently pull the vane apart so it splits, then stroke it back together - it "zips" shut! Those are the barbule hooks.

PREDICT before lunch: When we drip plain water on a feather (and on a paper towel) in the Hydro Lab, what will happen to each? What about after we add a drop of soap?

LEARNING OBJECTIVES

- Explain why water rolls off a duck's feathers.
- Identify key parts of a bird's feather.
- Define 'hydrophobic' in simple terms.

INSTRUCTIONS

Today, we're learning about why ducks stay dry! We've watched some videos and read about how their feathers work. Now, answer these questions to show what you've learned about feathers and how they repel water. Remember to think about both the feather's structure and the special oil ducks use.

WORKSHEETS & ACTIVITIES

Intro: Why Water Rolls Off a Duck (watch first) - Practice

Book: Spectrum Science Grade 5, pg 45-47

1. What two main things help water roll off a duck's feathers?

2. What is the name for the tiny hooks that zip a feather together?

3. Where does a duck get the oil it spreads on its feathers?

4. What does 'hydrophobic' mean?

5. What happens to water when it touches an oily, tightly-zipped feather?

ANSWER KEY (for adult use only)

1. Feather structure (like tiny hooks) and special oil.

2. Barbules.
3. From a special gland near its tail.
4. Water-fearing, or repelling water.
5. It forms beads and rolls off.

Block 4 of 7 · Reagan · 2026-06-18

5. Lunch + reset

12:00 PM · 60 min

Lunch and a calm reset before the afternoon.

LEARNING OBJECTIVES

- Eat a healthy lunch.
- Clean up the lunch area.
- Relax quietly for a short time.
- Be ready for the next activity.

INSTRUCTIONS

It's time for lunch and a short break! Please eat your lunch slowly and mindfully. After you finish, make sure to clean up your eating area completely. Then, find a quiet spot to relax, read a book, or just sit still for a few minutes before we start our afternoon work.

WORKSHEETS & ACTIVITIES

Lunch + reset - Practice

1. What is one healthy food you ate for lunch today?

2. What is one thing you did to clean up after lunch?

3. What did you do to relax during your reset time?

ANSWER KEY (for adult use only)

1. Answers will vary.
2. Answers will vary (e.g., put plate in sink, wiped table).
3. Answers will vary (e.g., read, sat quietly, drew).

Block 5 of 7 · Reagan · 2026-06-18

6. 3-Duck Measurement Adventure

1:00 PM · 40 min

Hands-on with our 3 live ducks! Measure each duck (length beak-to-tail, height, and weight if we can). Then fill the bathtub or a tub of water and explore volume/capacity: how many cups to fill it, convert cups to mL/L, and compare each duck's measurements. Practice converting length (in to cm) and volume (cups to liters) using real numbers from the ducks.

LEARNING OBJECTIVES

- Measure the length and height of three ducks.
- Practice converting inches to centimeters and cups to liters.
- Calculate the volume of a tub of water using cups and convert to liters.

INSTRUCTIONS

Today, we're going on a 3-Duck Measurement Adventure! First, carefully measure each duck's length (beak to tail) and height. Write down your measurements. Next, fill a tub with water and count how many cups it takes. Finally, use your measurements to answer the questions below.

WORKSHEETS & ACTIVITIES

3-Duck Measurement Adventure - Practice

Book: Math Measurement Fun!, pg 45-50

1. Duck 1 is 12 inches long. How many centimeters is that? (1 inch = 2.54 cm)
2. Duck 2 is 8 inches tall. How many centimeters is that? (1 inch = 2.54 cm)
3. It took 20 cups of water to fill the tub. How many liters is that? (1 cup = 0.24 liters)
4. If Duck 3 is 10 inches long, and Duck 1 is 12 inches long, what is the difference in their lengths in centimeters?

ANSWER KEY (for adult use only)

1. 30.48 cm
2. 20.32 cm
3. 4.8 liters
4. 5.08 cm

7. Duck Hydro Lab: Why Water Beads Off a Duck

1:40 PM · 30 min · Science

Science + birds! Explore water with the ducks: (1) Drip water on a feather vs. a paper towel and watch it bead/roll off - talk about how duck feathers are waterproof thanks to preen oil. (2) Sink-or-float in the water dish: predict, then test small safe objects. (3) Stir in a tiny drop of dish soap and watch water stop beading - that is why soap on feathers is bad. Draw what you saw and write 1 sentence on why ducks stay dry.

LEARNING OBJECTIVES

- Explain why ducks stay dry in water.
- Observe how water beads on a duck's feather.
- Understand the effect of soap on waterproof surfaces.

INSTRUCTIONS

Today, we will explore why ducks stay dry! First, you will drip water on a feather and a paper towel to see the difference. Next, you will test if small objects sink or float in a dish of water. Finally, you will add a tiny drop of dish soap to the water and observe what happens to the water beading. Remember to draw what you saw and write one sentence about why ducks stay dry.

WORKSHEETS & ACTIVITIES

Duck Hydro Lab: Why Water Beads Off a Duck - Practice

Book: Spectrum Science Grade 5, pg 22-25

1. What happened when you dripped water on the feather?

2. What happened when you dripped water on the paper towel?

3. Why do ducks stay dry?

4. What happened when you added soap to the water?

5. Why is soap bad for a duck's feathers?

ANSWER KEY (for adult use only)

1. The water beaded up and rolled off.
2. The water soaked in.
3. Ducks stay dry because their feathers have a special oil that makes them waterproof.
4. The water stopped beading and spread out.
5. Soap removes the special oil, so the feathers are no longer waterproof.